

Neel Asheshbhai Shah

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Education

Master of Science in Software Engineering <i>San Jose State University, GPA - 3.8/4.00</i>	Aug 2024 – May 2026 <i>San Jose, CA</i>
Bachelor of Technology in Computer Science <i>Nirma University</i>	Oct 2020 – Jun 2024 <i>Ahmedabad, India</i>

Skills

Programming	Python, JavaScript, TypeScript, Java, Golang, C/C++
Web Technologies	React, Next.js, Express.js, Node.js, Flask, Django, CSS, HTML
Cloud & DevOps	AWS, GCP, Terraform, Docker, Jenkins, Kubernetes, CI/CD pipelines, Ansible
Agentic AI	LangChain, LangGraph, AutoGen, LangSmith, Llama Index, Vector Databases
ML & DL	PyTorch, TensorFlow, NumPy, Pandas, Matplotlib, Seaborn
Databases	MongoDB, Redis, DynamoDB, PostgreSQL, AWS RDS, SQL, MySQL

Experience

Software Engineering Intern — Juniper Networks, Sunnyvale, US - Developed a full-stack agentic AI platform automating analysis of Customer Use Cases (CUCs), reducing manual research time for engineers and enabling faster, data-driven decision-making. - Designed and implemented a stateful, multi-agent workflow with LangGraph, where each agent specialized in distinct tasks allowing intent-based query routing and resolution. - Created a data pipeline to ingest and process over 3,000 CUCs and 50 network profiles, building a knowledge base with two-stage retrieval for semantic search and database-backed config storage. - Engineered a suite of specialized tools and utilities empowering agents to efficiently execute tasks within the multi-agent workflow.	May 2025 – Aug 2025
Software Engineering Intern — Crest Data, Ahmedabad, India - Deployed and managed microservices in an e-commerce application using Terraform, Docker, and Jenkins, ensuring 99.9% uptime. Provisioned cloud resources automatically with Terraform, reducing setup time from hours to minutes. Containerized microservice with Docker, maintaining consistent environments across dev, staging, and production. - Built a CI/CD pipeline with Jenkins to streamline build, test, and deployment, providing faster, more reliable releases.	Jun 2023 – Aug 2023

Projects

Nit – Custom Version Control System <i>C/C++, File I/O, SHA-1 Hashing</i> - Built Nit, a Git-like version control system from scratch in C/C++, implementing core commands init, add, commit, log, and status without relying on existing VCS libraries. - Designed a custom object storage model using SHA-1 content hashing to track file blobs and commit snapshots, enabling efficient deduplication and reliable change history retrieval. - Implemented a commit graph traversal engine to walk parent-child commit links, enabling accurate history reconstruction for the log command and laying the foundation for future branching and merge support.	
Distributed Video Encoding System <i>Python, gRPC, Raft consensus, FFmpeg, Docker, Kubernetes</i> - Implemented a fault-tolerant distributed video encoding pipeline using a master-worker architecture, supporting parallel execution across multiple machines with gRPC-based inter-process communication for task scheduling, status reporting, and failure recovery. - Implemented Raft-inspired leader election and heartbeat mechanisms to ensure high availability and consistent task coordination during master node failures in a distributed environment. - Optimized compute utilization by partitioning video streams into keyframe-aware segments using FFmpeg for efficient parallel processing.	GitHub
Collaborative Cloud IDE <i>MERN stack, Docker, Kubernetes, Websocket</i> - Designed and implemented an online collaborative coding platform (IDE) where multiple users collaborate on the same project in real time, supporting up to 100 concurrent users. - Provisioned dedicated pods per project to ensure isolated environments and applied Kubernetes ingress controllers to dynamically route project-specific requests. - Facilitated real-time collaboration by linking services via WebSocket, allowing users to edit and execute code simultaneously.	GitHub